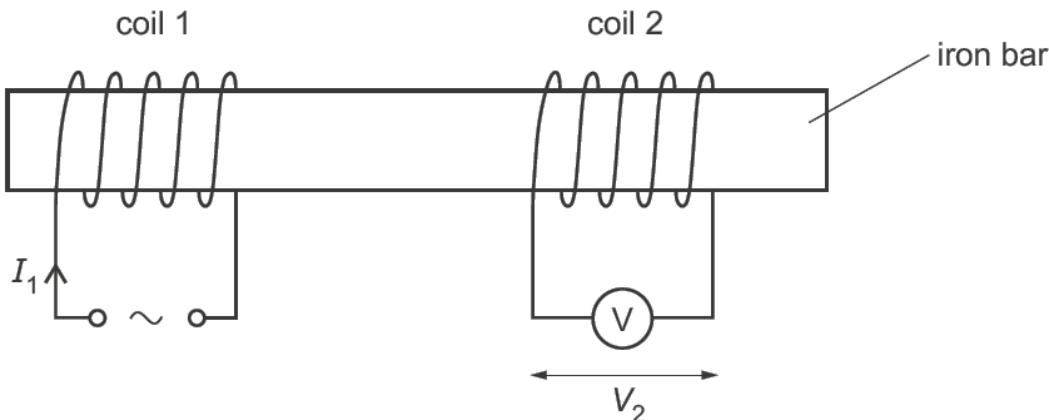


5	(a)	State Faraday's law of electromagnetic induction.
		
		
		 [2]
		Faraday's law of electromagnetic induction states that the <u>magnitude of the induced e.m.f. in a conductor [B1]</u> is <u>directly proportional to the rate of change of magnetic flux linkage [B1]</u> of the conductor.
	(b)	Two coils of insulated wire are wound on an iron bar, as shown in Fig. 5.1.  The diagram shows a horizontal iron bar with two coils wound around it. Coil 1, on the left, is connected to an AC power source. The current is labeled I_1 with an upward arrow. Coil 2, on the right, is connected to a voltmeter, represented by a circle with a 'V' inside. The voltmeter reading is indicated by a double-headed arrow below it, labeled V_2. The iron bar is labeled 'iron bar' with a line pointing to it. Fig. 5.1

There is a current I_1 in coil 1 that varies with time t as shown in Fig. 5.2.

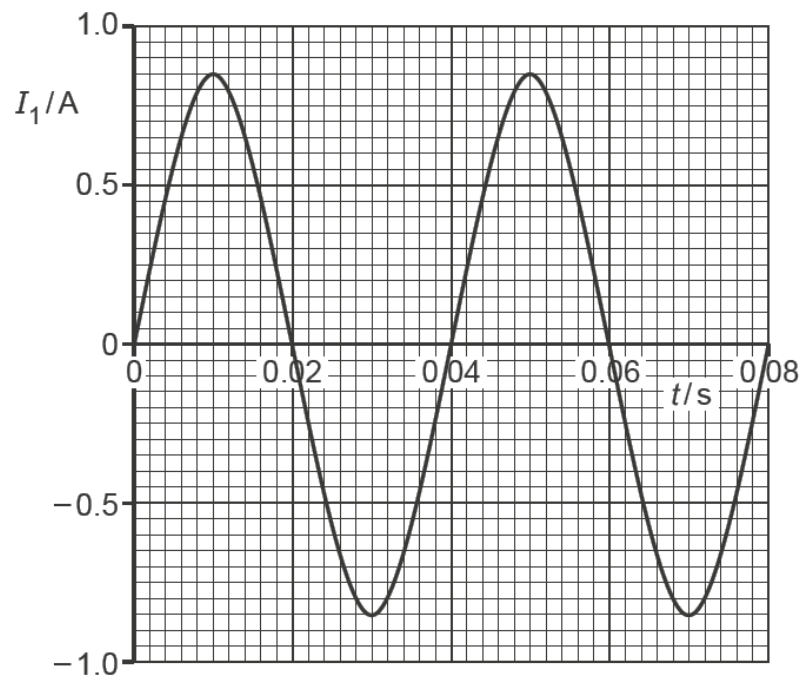


Fig. 5.2

- (i) The variation with t of I_1 can be represented by the equation

$$I_1 = A \sin B(t)$$

where A and B are constants.

Use Fig. 5.2 to determine the values of A and B . Give units to your answers.

$A = \dots\dots\dots$ unit $\dots\dots\dots$

$B = \dots\dots\dots$ unit $\dots\dots\dots$

[2]

From Fig. 5.2, $A = 0.85 \text{ A}$ [A1]

$$B = 2\pi / T = 2\pi / 0.040 \\ = 160 \text{ (or } 157) \text{ rad s}^{-1} \text{ [A1]}$$

(* Answers must include appropriate units)

- (ii) The current in coil 1 gives rise to a magnetic field with a flux density that is proportional to I_1 .

An electromotive force (e.m.f.) is induced across coil 2. The potential difference (p.d.) across coil 2 is measured using a voltmeter that gives a root-mean-square (r.m.s.) value of 4.6 V.

On Fig. 5.3, sketch a graph to show the variation with t of V_2 between $t = 0$ and $t = 0.08$ s.

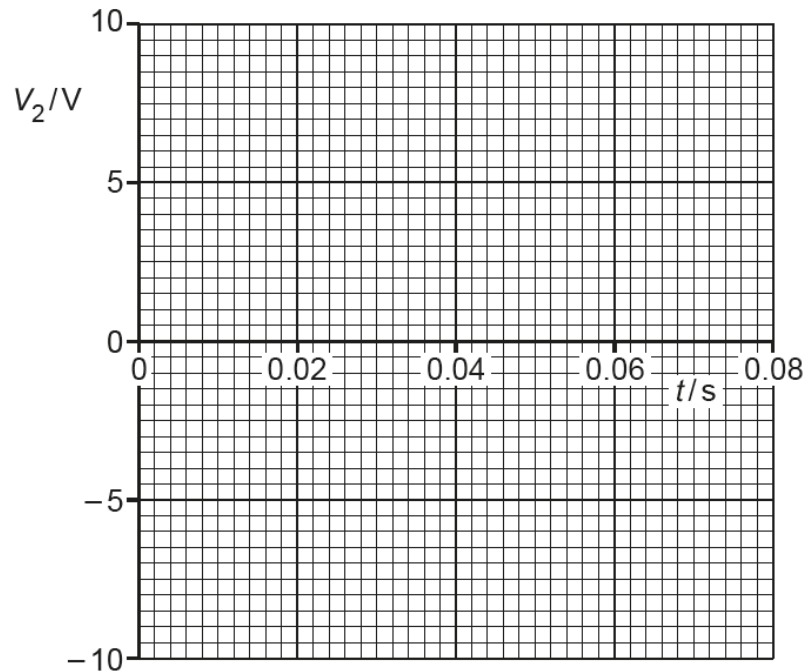


Fig. 5.3

[3]

[B1] – 2 cycles of a negative cosine graph (starting at (0, - 6.5)) with a

[B1] – Minimum e.m.f of zero at $t = 0.01, 0.03, 0.05$ and 0.07 s

- Maximum (magnitude) e.m.f. at $t = 0, 0.02, 0.04, 0.06$ and 0.08 s.
- period of 0.040 s.

[B1] - $V_0 = \sqrt{2}(4.6) = \underline{\underline{+/- 6.5 \text{ V at peak e.m.f. times.}}}$

- (iii) Use the laws of electromagnetic induction to explain the shape of your graph in (b)(ii).

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[3]

According to Faraday's law,

- Magnitude of V_2 is proportional to gradient of the $I_1 - t$ graph (Fig. 5.2)
- V_2 has maximum magnitude when gradient of the $I_1 - t$ graph is the steepest.

		- V_2 is <u>zero</u> when $I_1 - t$ graph is <u>horizontal momentarily</u> at its peak and minimum points. 2 x [B1] – Any 2 points above, 1 mark each According to Lenz's law, - The induced e.m.f produces effects that opposes the magnetic flux linkage that causes it. - Hence, <u>V_2 changes sign</u> when the <u>sign of the gradient</u> of the $I_1 - t$ graph <u>changes</u>. [B1] – 2nd point above
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[Total: 10]

